

Research papers

Effects of vegetation restoration on infiltration patterns and preferential flow in semi-arid areas with shallowly buried soft bedrock (Pisha sandstone) in China

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ABSTRACT

Vegetation restoration significantly influences ecohydrological processes in dryland, particularly in the Pisha sandstone area where shallow bedrock burial hinders infiltration. Despite its importance, the effects of vegetation restoration on infiltration patterns and preferential flow in such regions are poorly understood. This study employed dye tracer methods and image analysis technologies to evaluate the impact of sea buckthorn woodlands (3–4 years (Y3), 14 years (Y14)) and natural grassland (NG) on infiltration patterns and preferential flow in the Pisha sandstone area, with bare land (BL) as a control. The results showed that vegetation restoration enhanced the development of preferential flow by promoting root growth and improving soil properties. The degree of preferential flow followed the order: Y14 > Y3 > NG > BL. Roots and soil porosity were identified as the primary factors driving preferential flow development, contributing 71.0 % and 71.7 % to its variation, respectively. Increased root presence and pore structures further encouraged macropore flow within sea buckthorn woodland and natural grassland, especially in the Y14 plots. These results highlight the positive impact of vegetation restoration on soil preferential flow, with sea buckthorn woodland demonstrating superior improvement compared to natural grassland, especially at Y14. Given the critical role of preferential flow in soil water replenishment, it is recommended to prioritize planting sea buckthorn in the Pisha sandstone area characterized by shallow bedrock burial. These insights provide valuable guidance for addressing water limitations in arid and semi-arid regions worldwide facing similar challenges.

1. Introduction

Arid and semi-arid regions comprise 41 % of the world's land surface (Reynolds et al., 2007). These areas, heavily impacted by climate change and human activities, are prone to soil erosion and land degradation (Prävalie, 2021). Ecological restoration has proven effective in preventing soil degradation and enhancing biodiversity in arid regions (Jia et al., 2019; Zanini et al., 2021; Zhou et al., 2021). In many dryland environments, the soil layer is extremely thin (Lapides et al., 2024). Shallowly buried bedrock, with its dense pore structure and low permeability, restricts the vertical infiltration of rainwater (Qin et al., 2015; Rodríguez-Robles et al., 2015). This results in significant water

deficits, severely limiting sustainable reforestation efforts (Qin et al., 2015). However, vegetation roots can extract water from cracks in weathered bedrock, using this “underground reservoir” as a potential water source during dry periods (Lapides et al., 2024; Lin et al., 2024). Despite this potential, the feedback effects of vegetation restoration on hydrological processes in such areas are rarely discussed. Understanding the relationship between vegetation restoration and hydrological processes can help manage water resources better and promote sustainable vegetation restoration in arid and semi-arid regions worldwide.

Soil water is a vital resource in arid and semi-arid regions worldwide, where rainfall infiltration sustains water replenishment (Jia et al., 2017; Liu et al., 2023). There are two main types of water infiltration patterns:

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matrix infiltration and preferential infiltration (Stumpp and Maloszewski, 2010). The former characterizes the gradual downward movement of water within the soil matrix uniformly (Alaoui et al., 2011), while the latter indicates rapid and uneven movement along specific pathways (Bargués Tobella et al., 2014; Zhang et al., 2016). Preferential flow paths typically form due to macropores in the soil, influenced by factors such as soil animal activity (Murphy and Banfield, 1978), root growth and decay (Cui et al., 2019; Guo et al., 2019), cracks from dry-wet cycles or freeze-thaw processes (Bracken et al., 2013), and gaps between soil and rocks (Sohrt et al., 2014). Additionally, heterogeneous infiltration, unstable wetting fronts, water repellency in sandy soil, and varying hydraulic properties across different soils or soil-rock interfaces contribute to preferential flows, such as unsteady flows (including finger flows) and funnel flows (Babel et al., 1995; Nimmo, 2021). Though the volume of preferential flow paths in soil pores is relatively small, the water moving through these paths significantly impacts the overall infiltration process, contributing between 15 % and 85 % of the total infiltration (Laine-Kaulio et al., 2014; Zhang et al., 2021). Therefore, understanding water infiltration patterns and preferential flow is essential for managing dryland rainfall infiltration, water distribution, and groundwater replenishment (Beven and Germann, 2013; Holman et al., 2004). Although accurately quantifying these spatially variable preferential flows is challenging (Allaire et al., 2009; Cey and Rudolph, 2009), integrating dye tracer methods with image analysis techniques offers an effective and straightforward solution (Kodesová et al., 2012; Mooney and Morris, 2008).

Vegetation restoration significantly impacts soil water infiltration patterns and the development of preferential flow, thereby playing a crucial role in enhancing ecosystem service functions (Barral et al., 2015). Wang et al. (2024) observed that funnel flow and finger flow are the primary flow behaviors in artificial forests and natural secondary forests, respectively, within ancient periglacial landforms. Moreover, Qiu et al. (2023) noted that the presence of more roots and soil macropores in woodlands, shrublands, and grasslands leads to deeper water infiltration in hilly and gully areas on the Chinese Loess Plateau. Similarly, Guan et al. (2024) found that forest soils exhibit a more pronounced preferential flow phenomenon compared to grasslands, confirming that vegetation restoration affects infiltration patterns on the Chinese Loess Plateau. Several studies have also shown that vegetation alters infiltration patterns and preferential flow by changing soil physical and chemical properties such as porosity, organic matter content, and aggregate stability (Bargués Tobella et al., 2014; Kan et al., 2019; Mei et al., 2018; Qiu et al., 2023; Sun et al., 2018). Roots play a pivotal role in this process by increasing soil organic matter levels through secretions and root decomposition, thus enhancing soil porosity (Fageria and Stone, 2006; Vannoppen et al., 2015). Additionally, both root channels around living roots and those formed after root decay facilitate preferential water flow (Backnäs et al., 2012; Mitchell et al., 1995; Nimmo, 2021). While these studies highlight the importance of vegetation in altering soil water flow behavior, research on semi-arid regions with shallowly buried soft bedrock remains limited. Further investigation is needed to fully understand these processes in such environments.

The Pisha sandstone area, located in the semi-arid region of northern China, is recognized as “the most severe area of soil erosion in the world” (Liang et al., 2019), with soil erosion rate surpassing 20000 t/(km²·a) (Ma and Zhang, 2016; Zhang et al., 2009). This region’s soil layer is extremely thin, with bedrock either shallowly buried or exposed on the surface (Yang et al., 2023a). Pisha sandstone, a special soft rock with low erosion resistance, rapidly decomposes into sand or mud under wind or water erosion, hindering local vegetation growth (Liu et al., 2023). The Pisha sandstone contains a high proportion of montmorillonite (30 %), a clay mineral (Ma and Zhang, 2016). When exposed to water, montmorillonite expands, causing the internal structure of the sandstone to collapse (Liang et al., 2019; Shi et al., 2009; Ye et al., 2006). The clay particles released during this process can block soil pores, further impeding water infiltration (Ma and Zhang, 2016). The transition layer

between soil and Pisha sandstone further impedes water infiltration in this area, leading to a stable infiltration rate of less than 0.3 mm·min⁻¹ (Dong et al., 2022). Consequently, low permeability and severe soil erosion have led to scarce natural vegetation and a fragile ecological balance in the region. Erosion management efforts in the Pisha sandstone area began in the 1950 s (Liang et al., 2019). Sea buckthorn (*Hippophae rhamnoides*) has been identified as a “pioneer species” for controlling soil and water loss due to its remarkable physiological characteristics, including tolerance to cold, drought, and poor soil, nitrogen fixation ability, and strong cloning and reproduction capabilities (Liu et al., 2022). Decades of dedicated vegetation restoration efforts have resulted in the development of two primary vegetation cover types: natural grassland and sea buckthorn woodland. Vegetation plays a critical role in mitigating soil erosion, improving soil quality, and maintaining the hydrological cycle. However, it remains uncertain whether vegetation restoration can profoundly affect the infiltration process in the Pisha sandstone area with shallow bedrock burial, especially regarding infiltration patterns and preferential flow. Further research is needed to understand these effects comprehensively.

In the study, two typical vegetation restoration types (sea buckthorn woodland of different ages and natural grassland) in the Pisha sandstone area were selected, with bare land used as a control, to analyze water infiltration patterns and preferential flow phenomena. The specific objectives of this study were (1) to investigate the differences in infiltration patterns and preferential flow under these vegetation types, and (2) to identify the main factors affecting the development of preferential flow in the Pisha sandstone area.

2. Materials and methods

2.1. Study site

The study was conducted in the Geqiugou basin, located in Junge Banner, Ordos City, Inner Mongolia Autonomous Region, China (110°25′–110°48′ E, 39°42′–39°50′ N) (Fig. 1). This basin, situated in the heart of the exposed Pisha sandstone area, features undulating terrain with an average altitude ranging from 820 to 1528 m and covering a total area of 96 km². The region experiences a semi-arid continental monsoon climate within the middle temperate zone. The average annual rainfall is approximately 441.4 mm (2008–2022) (Yang et al., 2024), mainly occurring from June to September, while annual evaporation averages around 2093 mm. Temperatures typically range from 6.2 to 8.7 °C. The area is characterized by four distinct seasons, strong solar radiation, and prolonged sunshine, with an average annual sunshine duration of over 2900 h. Spring is marked by strong and prolonged winds, with an average annual wind speed of 3.2 m·s⁻¹, peaking at 32 m·s⁻¹, and 10 to 30 windy days annually (Liu et al., 2023). The Pisha sandstone in the basin dates back to the Cretaceous and Jurassic. The dominant soil types include chestnut soil, loess, and aeolian sand soil (Table 1). The primary herbaceous vegetation consists of *Leymus chinensis* and *Artemisia ordosica*. Prominent artificial vegetation includes *Hippophae rhamnoides*, *Caragana korshinskii*, *Pinus tabulaeformis*, and *Prunus sibirica*.

2.2. Experimental design

Extensive areas of sea buckthorn woodland and natural grassland were identified in the basin, offering an opportunity to study the effects of vegetation restoration on soil infiltration patterns and preferential flow. In August 2022, after consulting with local forestry bureau personnel and permanent residents, we selected two types of vegetation restoration: sea buckthorn woodland of different forest ages and natural grassland (NG), with bare land (BL) serving as a control (Fig. 1). To address soil erosion, extensive sea buckthorn plantings were undertaken in 2008 and again in 2018–2019, resulting in sea buckthorn woodlands aged 14 years (Y14) and 3–4 years (Y3). The planting methods were

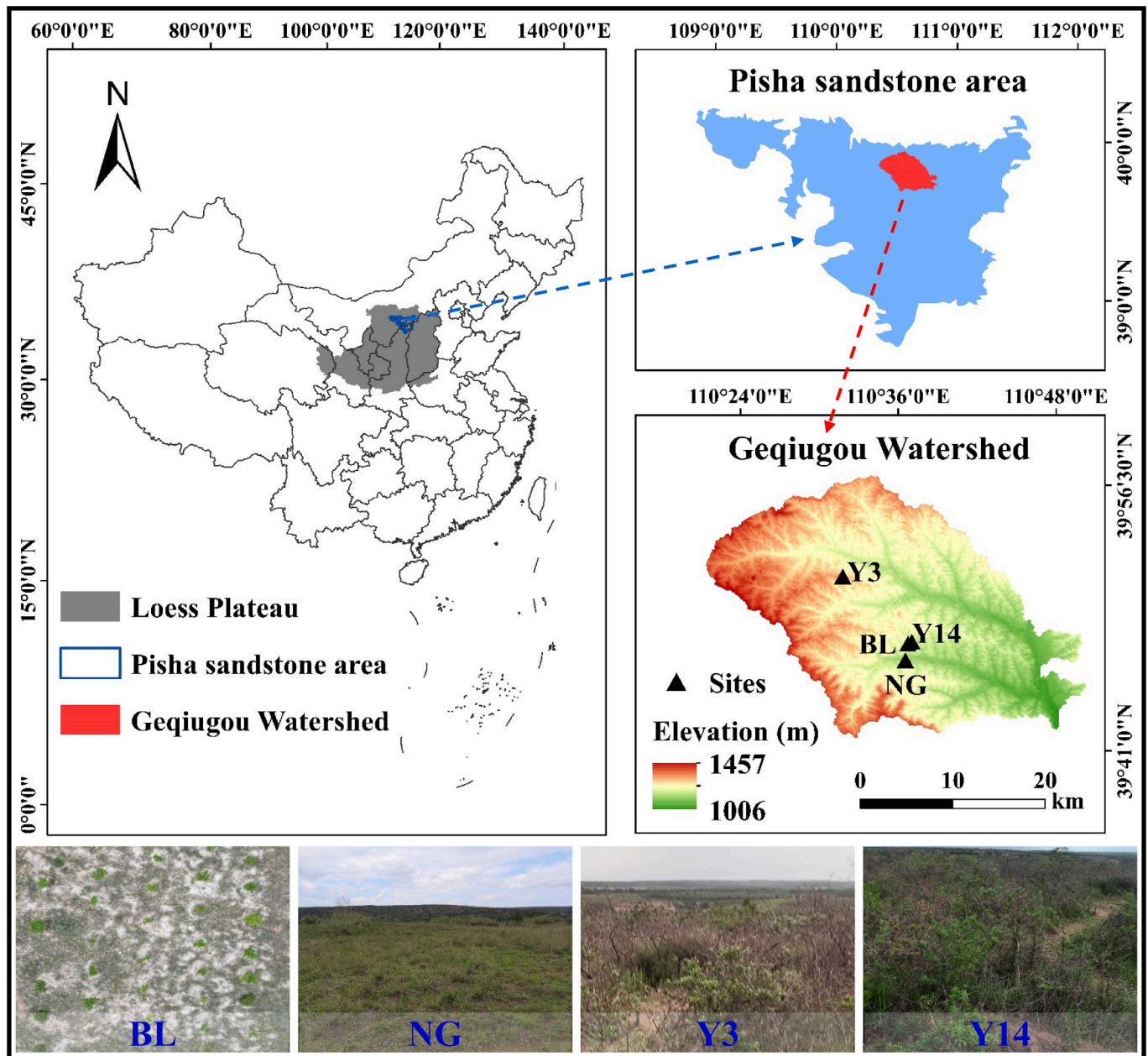


Fig. 1. Geographic location of study area. BL, bare land. NG, natural grassland. Y3, sea buckthorn woodland aged 3–4 years. Y14, sea buckthorn woodland aged 14 years.

consistent, allowing for natural growth post-planting. However, due to severe soil erosion and limited water and nutrients, bare land was predominantly covered with sparse herbaceous vegetation. With small differences in elevation and slope ($<6^\circ$) across all plots (Table 1), topographic influences were considered negligible. Subsequently, three $1\text{ m} \times 1\text{ m}$ quadrats were randomly selected in each plot for dye tracer experiments (Fig. S1a). Three repeated soil and vegetation samples were collected from each plot.

2.2.1. Dye tracer experiment

Aboveground herbaceous vegetation and litter were collected from the quadrat area, followed by levelling the land to eliminate slope-related effects on infiltration. There was no rainfall for a week before the experiment. An iron frame measuring 50 cm in length, 50 cm in width, and 40 cm in height was inserted into the soil to a depth of 5 cm (Fig. S1b). The soil around the frame was gently compacted using a

rubber mallet to prevent water seepage. In this experiment, we simulated the local maximum daily rainfall (60 mm) (Yang et al., 2023b). The 15 L of brilliant blue solution ($4\text{ g}\cdot\text{L}^{-1}$) was sprayed manually onto the soil surface within an iron frame at a rate not exceeding $20\text{ mm}\cdot\text{h}^{-1}$ to prevent water accumulation. The iron frame was then covered with a waterproof cloth to minimize evaporation and additional rainwater infiltration. After 24 h of infiltration, the iron frame was carefully removed to avoid disturbing the surrounding soil. Vertical soil profiles were dug to the depth of the deepest stain, and one dyed section was collected from the center of each quadrat (Fig. S1c), totaling 12 dyed sections. To mitigate edge effects, the dyed section located 5 cm from the iron frame was excluded, resulting in dyed sections that were 40 cm wide. These dyed sections were then used for subsequent image analysis.

2.2.2. Measurement of vegetation parameters

The aboveground herbaceous vegetation was oven-dried at 60°C for

Table 1

Basic characteristics of bare land (BL), natural grassland (NG), and sea buckthorn woodlands (Y3 and Y14) in Pisha sandstone area.

Plot	Altitude (m)	Slope (°)	Sand (%)	Silt (%)	Clay (%)	HB (g·m ⁻²)	LC (g·m ⁻²)
BL	1192	5	88.09 ± 0.56a	9.84 ± 0.49ab	2.07 ± 0.09b	52.35 ± 10.10c	47.87 ± 10.04c
NG	1153	3	90.30 ± 0.77a	7.08 ± 0.28b	2.62 ± 0.60b	70.73 ± 6.02bc	113.06 ± 2.32b
Y3	1289	6	89.13 ± 2.43a	7.98 ± 1.60b	2.89 ± 0.83b	87.87 ± 5.46b	119.60 ± 7.84b
Y14	1192	5	82.00 ± 2.52b	12.87 ± 1.85a	5.13 ± 0.69a	156.29 ± 0.73a	234.07 ± 9.39a

Notes: BL, bare land. NG, natural grassland. Y3, sea buckthorn woodland aged 3–4 years. Y14, sea buckthorn woodland aged 14 years. HB, aboveground herbs biomass. LC, litter accumulation. Different lowercase letters represent a significant difference at 0.05 levels ($P < 0.05$) among different sites. Values are means ± standard error (n = 3 replications).

one week, while the litter was oven-dried at 60 °C for 24 h. After drying, aboveground herbs biomass (HB, g·m⁻²) and litter accumulation (LC, g·m⁻²) were calculated. The staining depth of all squares was less than 40 cm, so root samples were collected at 10 cm intervals up to this depth. The roots were scanned using an Epson Perfection V750 Pro root scanner. Win RHIZO software was used to determine root trait parameters, including root average diameter (RAD, mm), root length density (RLD, mm·cm⁻³), and root volume density (RVD, mm³·cm⁻³). The vegetation roots were then dried in an oven at 60 °C until they reached a constant weight, which was used to calculate the root biomass (RB, g·cm⁻³).

2.2.3. Measurement of soil properties

Samples were collected from three soil layers (0–10 cm, 10–20 cm, and 20–40 cm). Cutting rings (100 cm³) were used to gather undisturbed soil samples for determine soil bulk density (BD, g·cm⁻³) and total porosity (TP, %). Three undisturbed soil samples, each collected near the quadrat using a 5 cm diameter soil drill, were combined into a single composite sample (Fig. S1). This process was repeated for each soil layer in every plot, resulting in three composite samples per layer. These samples were then used to analyze soil particle composition and organic matter content. Soil organic matter (SOM, g·kg⁻¹) was measured using the dichromate titration-external heating method. The Master 2000 laser particle size analyzer was utilized to evaluate soil particle composition. Additionally, undisturbed soil lumps weighing at least 250 g were collected, air-dried indoors, and then subjected to the wet screen method to measure the mean weight diameter of water-stable aggregates (MWD, mm). The calculation formula is as follows:

$$MWD = \sum_{i=1}^n R_i W_i \quad (1)$$

where R_i is the i th a graded average diameter (mm), W_i is the weight of the i th a graded levels.

2.3. Quantification of preferential flow

Photoshop CS6 software was utilized to adjust the size, color levels, and exposure degree of the dyed images. This software was also used to replace colors in the images and convert them into binary images. To further process these images, Image-Pro Plus software was employed to reduce noise and analyze the bitmap of the binary images, deriving a binary digital matrix. All dyed images were 40 cm wide and 40 cm high. These dyed image were divided into 40 rectangular grids of 40 cm² to calculate the preferential flow characteristic parameters corresponding to each grid (40 cm wide by 1 cm high). Specific dyed image analysis and

quantification procedures were shown in Fig. S3.

2.3.1. Characteristic parameter of preferential flow

Dye coverage (DC, %) is the proportion of the dyed area to the total soil profile (Flury et al., 1994).

$$DC = \left(\frac{D}{(D + ND)} \right) \times 100 \quad (2)$$

Where D is the dyed area of the soil profile (cm²) and ND is the undyed area of the soil profile (cm²).

Uniform infiltration depth (UniFr, cm) is the soil depth corresponding to a dye coverage greater than 80 % (Van Schaik, 2009).

Preferential flow fraction (PF_{fr}, %) represents the percentage of total infiltration through the preferential flow channels (Van Schaik, 2009).

$$PF_{fr} = \left(1 - \frac{UniFr \times W}{TSA} \right) \times 100 \quad (3)$$

Where W is the width of the dyed section (here 40 cm) and TSA is the dyed area of the soil profile (cm²).

As the degree of soil preferential flow increases, the preferential flow fraction also rises, while the values for dye coverage and uniform infiltration depth decrease accordingly (Zhang et al., 2018).

2.3.2. Comprehensive evaluation index of preferential flow

The multi-index evaluation method was employed to comprehensively assess the development degree of preferential flow (Guan et al., 2023; Wu et al., 2022). Initially, various characteristic parameters of preferential flow (DC, UniFr, and PF_{fr}) were standardized. For parameters positively correlated with preferential flow development, we used Formula (4) for computation, while for those negatively correlated, we applied Formula (5). Next, the mean and mean square errors for each standardized parameter were computed (Formulas (6) and (7)). The weights for each parameter were then determined using the mean square error decision approach (Formula (8)). Finally, the preferential flow index (PFI) was derived based on the respective weights assigned to each preferential flow parameter (Formula (9)).

$$Z_{ij} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad (4)$$

$$Z_{ij} = \frac{X_{\max} - X_i}{X_{\max} - X_{\min}} \quad (5)$$

$$E(G_j) = \frac{1}{m} \sum_{i=1}^m Z_{ij} \quad (6)$$

$$\sigma(G_j) = \sqrt{\frac{1}{m} \sum_{i=1}^m (Z_{ij} - E(G_j))^2} \quad (7)$$

$$W(G_j) = \frac{\sigma(G_j)}{\sum_j \sigma(G_j)} \quad (8)$$

$$PFI = \sum_{j=1}^n Z_{ij} W(G_j) \quad (9)$$

Where i is the ordinal number of sites; j is the ordinal number of parameters; m is the total number of sites, and n is the total number of parameters; X_i is the actual value of parameter of the i th site; $X_{\min}$ is the actual minimum value of the parameter; $X_{\max}$ is the actual maximum value of the parameter; Z_{ij} is the standardized value of the parameter; G_j is the j th parameter, such as DC, UniFr, and PF_{fr}; $E(G_j)$ is the mean value of the j th parameter; $\sigma(G_j)$ is the mean square error of the j th parameter; $W(G_j)$ is weight coefficient of the j th parameter; and PFI is the comprehensive evaluation index of preferential flow.

2.4. Date analysis

One-way analysis of variance (ANOVA) with Duncan's post hoc test was applied to assess the significant differences among variables across different sites ($P < 0.05$). Person's correlation analysis and linear regression analysis were used to determine the relationship between variables. Redundancy analysis (RDA), environmental importance test, and variance partitioning analysis (VPA) were conducted using the "vegan" package in R. Partial least squares path modelling (PLS-PM) was performed with the "plsmpm" package in R. All statistical analyses were carried out using R version 4.2.2. Table 2 showed the description of the most abbreviations used in this study.

3. Results

3.1. Root traits

The RAD of BL, NG, and Y3 gradually decreased with soil depth, while the RAD of Y14 first increased and then decreased, peaking in the 10–20 cm layer (5.71 mm) (Fig. 2). Additionally, the RLD, RVD, and RB of NG, Y3, and Y14 were greater than those of BL across all soil layers. Notably, Y14 exhibited significantly higher RAD, RLD, RVD, and RB compared to BL, with increases of 1.03 times, 4.16 times, 6.41 times, and 13.28 times (mean value), respectively ($P < 0.05$). Moreover, there was no significant difference in root traits between Y3 and NG ($P > 0.05$), although the root trait parameters of Y3 were slightly higher than those of NG (except RLD). Furthermore, in comparison to those of NG, the RAD, RLD, RVD, and RB of Y14 showed significant increases of 74.59 %, 40.71 %, 101.54 %, and 234.84 % (mean value), respectively ($P < 0.05$). Therefore, vegetation restoration significantly improved root distribution in Pisha sandstone area, with sea buckthorn woodland demonstrating denser root growth than NG, particularly at the Y14 site.

3.2. Soil physical and chemical properties

The BD of NG, Y3, and Y14 in all soil layers was lower than that of BL (Fig. 3). Specifically, the BD of NG, Y3, and Y14 significantly decreased

by 6.97 %, 13.33 %, and 18.06 % (mean value), respectively, compared with that of BL ($P < 0.05$). The TP showed an opposite trend to BD. Additionally, the SOM of Y3 and Y14 was higher than that of BL across all soil layers, while the SOM of NG exceeded that of BL only in 10–20 cm soil layer. The SOM of NG and Y3 increased by 8.73 % and 30.90 %, respectively, compared with that of BL ($P > 0.05$). In contrast, the SOM of Y14 significantly increased by 83.34 % compared with that of BL ($P < 0.05$). Notably, the MWD of Y3 and Y14 was higher than that of BL only in 0–20 cm soil layer, while the MWD of NG was lower than that of BL across all soil layers. Therefore, vegetation restoration improved the soil porosity and structure while also increasing soil organic matter levels in the Pisha sandstone area. The enhancement was more pronounced in sea buckthorn woodland compared to natural grassland, with the improvement becoming more significant as the forest aged.

3.3. Infiltration patterns and preferential flow development degree

A representative dyed image was selected from each plot to analyze infiltration patterns. BL predominantly exhibited matrix flow infiltration (Fig. 4a and Fig. S2). In contrast, NG, Y3, and Y14 showed a large number of root holes and macropores in their stained profiles. Although the two types of pores differ in their morphological characteristics, size distribution, and material composition, clear traces of bright blue dye were observed around and within them, providing direct evidence that these pores guide water flow in the soil. Therefore, the infiltration pattern in sea buckthorn woodland and natural grassland was dominated by "macroporous flow". In addition, the water flow paths in the NG and Y3 were several and narrow, exhibiting a distinct "finger-like" shape. In contrast, the Y14 featured fewer but wider flow paths, displaying a clear "funnel" shape. Notably, sedimentary soft rock (Pisha sandstone) was identified at a depth of 20 cm in Y14 (Fig. S2). This sedimentary rock alternates irregularly with the soil, lacking a consistent pattern. The flow direction in Y14 is inclined and follows the structure of the sedimentary sandstone, resembling a "funnel". Furthermore, this variation may also be attributed to water moving through macroporous channels and subsequently dispersing into the surrounding soil matrix.

In all plots, the DC above the UniFr varied uniformly with soil depth, exceeding 80 % (Fig. 4b). However, below the UniFr, the DC of NG, Y3, and Y14 showed a more violent fluctuation trend with the increase of depth than that of BL, indicating greater instability of water infiltration in these plots. Additionally, NG, Y3, and Y14 exhibited lower DC and UniFr but larger PF_{fr} compared to BL (Fig. 5). Y14 demonstrated the lowest DC and UniFr, yet the highest PF_{fr} . Specifically, the DC and UniFr of Y14 significantly decreased by 20.19 % and 36.76 %, respectively, compared with those of BL ($P < 0.05$). The PF_{fr} of Y14 significantly increased by 4.23 times compared with that of BL ($P < 0.05$). Notably, the PFI for BL, NG, Y3, and Y14 were 0.03, 0.43, 0.70, and 0.90, respectively (Table 3). Therefore, preferential flow was most developed in sea buckthorn woodland (Y3 and Y14), followed by NG, and least in BL. As sea buckthorn aged, preferential flow further improved.

3.4. Relationships between soil properties, root traits, and preferential flow parameters

The root traits (RAD, RLD, RVD, and RB) were significantly negatively correlated with BD ($P < 0.01$ and $P < 0.001$) and significantly positively correlated with SOM ($P < 0.05$ and $P < 0.01$) (Fig. 6a). However, these root traits had no significant correlation with MWD ($P > 0.05$). This indicated that root traits significantly influenced soil porosity and nutrient levels. Additionally, DC was significantly negatively correlated with PF_{fr} ($P < 0.05$). PF_{fr} was significantly negatively correlated with BD ($P < 0.01$) and significantly positively correlated with both SOM and root traits ($P < 0.05$ and $P < 0.01$). However, PF_{fr} had no significant correlation with MWD ($P > 0.05$). Therefore, root traits, soil porosity, and nutrient levels significantly influenced the development of

Table 2
Specification of the abbreviations used in this study.

Abbreviations	Description	Unit
BL	Bare land used as a control in the experiment	
NG	Natural grassland	
Y3	Sea buckthorn woodland aged 3–4 years	
Y14	Sea buckthorn woodland aged 14 years	
Sand	Sand content (by volume percentage)	%
Silt	Silt content (by volume percentage)	%
Clay	Clay content (by volume percentage)	%
HB	Aboveground herbs biomass	g·m ⁻²
LC	Litter accumulation	g·m ⁻²
RAD	Root average diameter	mm
RLD	Root length density	mm·cm ⁻³
RVD	Root volume density	mm ³ ·cm ⁻³
RB	Root biomass	g·cm ⁻³
BD	Soil bulk density	g·cm ⁻³
TP	Total porosity	%
SOM	Soil organic matter content	g·kg ⁻¹
MWD	Mean weight diameter of water-stable aggregates	mm
DC	Dye coverage, the proportion of the dyed area to the total soil profile	%
UniFr	Uniform infiltration depth, the soil depth corresponding to a dye coverage greater than 80 %	cm
PF_{fr}	Preferential flow fraction, the percentage of total infiltration through the preferential flow channels	%
$E(G_j)$	Mean value of the j_{th} parameter	
$\sigma(G_j)$	Mean square error of the j_{th} parameter	
$W(G_j)$	Weight coefficient of the j_{th} parameter	
PFI	Preferential flow index, comprehensive evaluation index of preferential flow	

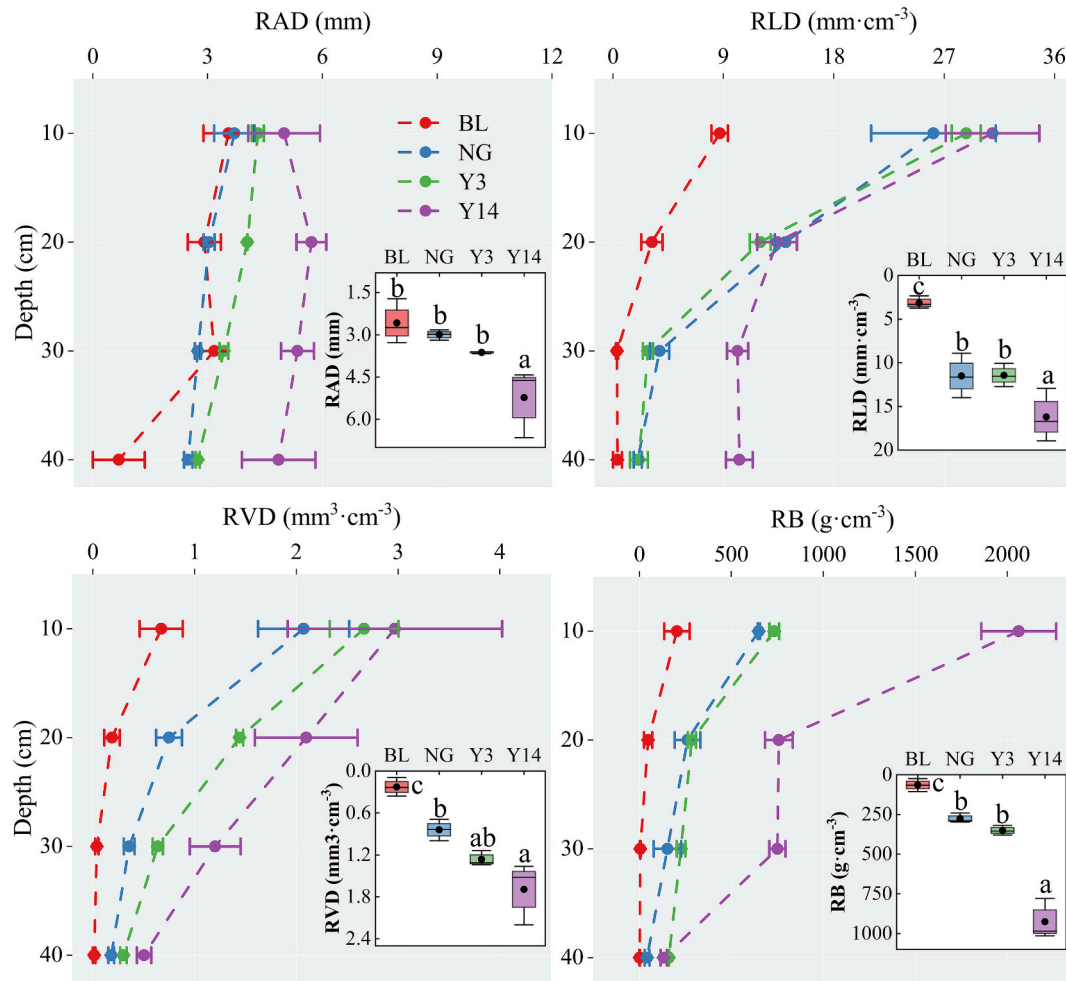


Fig. 2. Root traits in bare land (BL), natural grassland (NG) and sea-buckthorn woodland (Y3 and Y14). RAD, root average diameter. RLD, root length density. RVD, root volume density. RB, root biomass. The error bar represents the standard error ($n = 3$ replications). Different lowercase letters represent a significant difference at 0.05 levels ($P < 0.05$) among different sites.

preferential flow.

RDA1 and RDA2 explained 73.35 % and 0.94 %, respectively, of the variation in preferential flow (Fig. 6b). Among the factors, RB had the highest explanatory power, accounting for 78.76 % of the variation ($P < 0.01$) (Fig. 6c). BD explained 71.11 % ($P < 0.01$) of the variation in preferential flow. In contrast, SOM had the lowest explanatory power, accounting for only 34.67 % ($P > 0.05$) of the variation. Further analysis showed that soil porosity (BD, TP), root traits (RAD, RLD, RVD, and RB), and soil organic matter (SOM) explained 71.7 %, 71.00 %, and 38.2 % of the variation in preferential flow, respectively (Fig. 6d). Collectively, these factors accounted for 26.0 % of the variation. Therefore, soil porosity and root traits contributed more to development of preferential flow, followed by soil organic matter.

The constructed PLS-PM model explained 74 % of the variability in preferential flow (Fig. 7a). Soil porosity had the most significant positive total effect on preferential flow development (0.86), followed closely by vegetation roots (0.83) (Fig. 7b). Notably, vegetation roots directly influenced soil porosity, thereby indirectly promoting the degree of preferential flow.

4. Discussions

4.1. Effects of vegetation restoration on root distribution and soil properties

Our results showed that the RAD, RLD, RVD, and RB of sea buckthorn

woodland and natural grassland were higher than those of bare land (Fig. 2). This indicated that vegetation establishment improves the sparse and barren root distribution in the Pisha sandstone area. However, these root traits differed between sea buckthorn woodland and natural grassland due to varying recovery periods of sea buckthorn. In the short term, there was no significant difference in root traits between the NG and Y3. As restoration progressed, the return of litter to soil and nitrogen fixation of sea buckthorn enhanced soil nutrient status (Guo et al., 2010; Wang et al., 2015). This improvement in soil quality promotes root growth (Li et al., 2016), resulting in higher root trait parameters of Y14 than that of NG.

In turn, the secretions of living roots and the decomposed rotten roots further promote the accumulation of soil organic matter (Carrasco et al., 2006; Selvaraj et al., 2017; Yang et al., 2021). This was confirmed by the significant positive correlation between the root biomass and soil organic matter in this study (Fig. 6a and Fig. 7a). Therefore, higher root biomass and litter accumulation in sea buckthorn woodland have resulted in more soil organic matter compared to bare land (Fig. 3). The bonding effect of organic matter and the intertwining of vegetation roots further enhance soil particle aggregation (Xia et al., 2019; Zhai et al., 2020). However, our study found an insignificant positive correlation between soil organic matter (or root traits) and mean weight diameter of water-stable aggregates (Fig. 6a and Fig. 7a). This could be due to the lower organic matter content (or heterogeneity of root distribution). Previous research has shown that a lower soil organic carbon content ($< 15 \text{ g} \cdot \text{kg}^{-1}$) have little effect on aggregate stability (Le Bissonnais et al.,

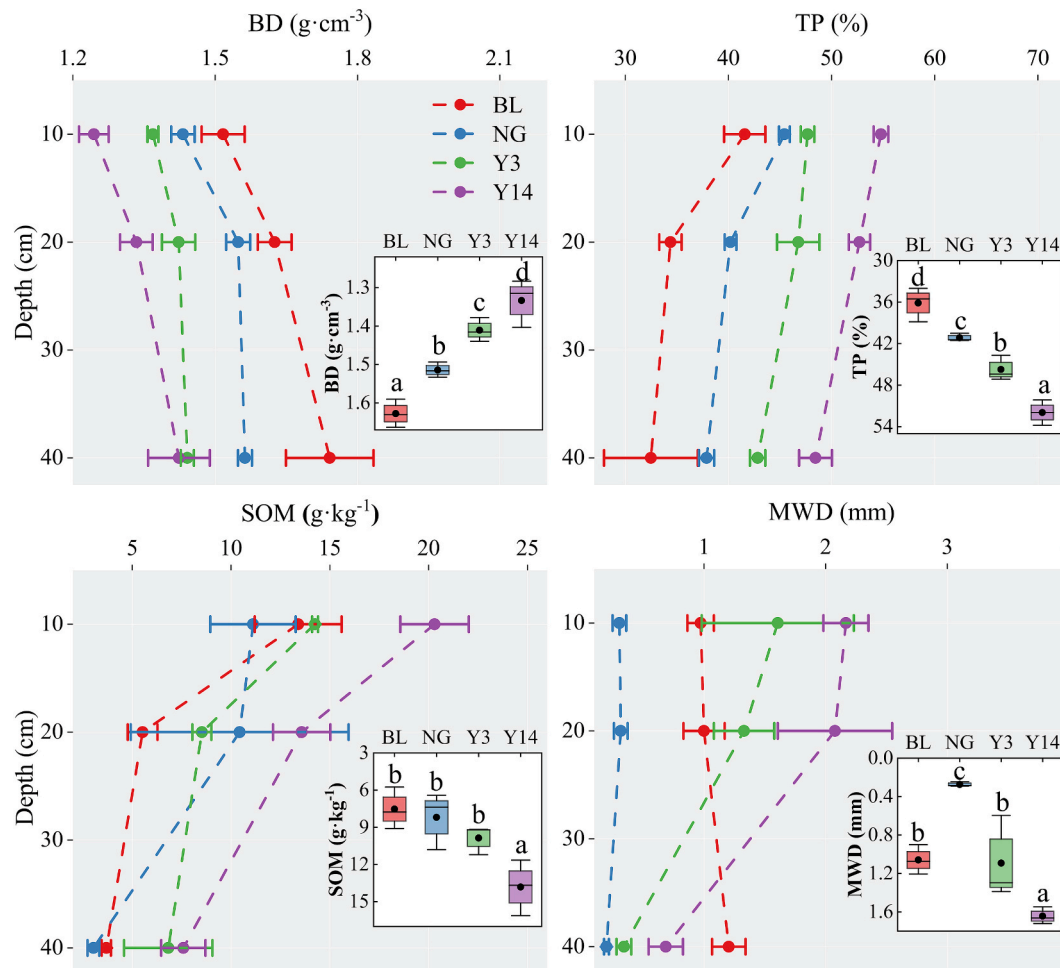


Fig. 3. Soil properties in bare land (BL), natural grassland (NG), and sea buckthorn woodland (Y3 and Y14). BD, bulk density. TP, total porosity. SOM, soil organic matter. MWD, soil mean weight diameter. The error bar represents the standard error ($n = 3$ replications). Different lowercase letters represent a significant difference at 0.05 levels ($P < 0.05$) among different sites.

2007). Indeed, in this study, all sites had the organic matter content below this threshold (Fig. 3). Despite this, the positive effects of organic matter or root traits on soil structure should not be ignored, as a good soil structure tends to result in higher soil porosity (Xia et al., 2017).

Notably, increasing evidence indicate that vegetation roots play a crucial role in improving soil porosity and reducing soil bulk density (Bodner et al., 2014). Greater root density leads to higher soil porosity (Hu et al., 2019), primarily because both root growth and decay could improve soil pore structure (Bowker et al., 2013; Wu et al., 2021). This study also confirmed this with significant correlations between root traits and soil bulk density (or total porosity) (Fig. 6a and Fig. 7a). Therefore, the soil porosity of sea buckthorn woodland and natural grassland was significantly higher than that of bare land (Fig. 3). In addition, the sea buckthorn woodland showed more root growth than natural grassland, resulting in better soil pore structure, especially at site Y14. Overall, vegetation restoration in Pisha sandstone area effectively improved soil properties. Sea buckthorn woodland demonstrated better restoration effects than natural grassland, with these effects increasing as the forest ages.

4.2. Effects of vegetation restoration on infiltration patterns and preferential flow

Macropore flow was observed in both sea buckthorn woodland and natural grassland (Fig. 4a). This could be attributed to the higher density of vegetation roots and soil pores in these plots compared to bare land

(Fig. 2, Fig. 3, and Fig. 4a). Sea buckthorn woodland exhibited thicker roots, deeper root distribution, and more root holes than natural grassland (Fig. 2, Fig. 3, and Fig. 4a), especially at site Y14. This indicated a more pronounced preferential flow in sea buckthorn woodland, aligning with previous studies (Guan et al., 2024). The growth and decay of vegetation roots significantly alter soil structure, which is key factor driving preferential flow (Cai et al., 2024). Previous studies have confirmed that root channels formed after root contraction or decomposition induce preferential water flow (Mitchell et al., 1995; Newman et al., 2004; Nobel and Cui, 1992; Noguchi et al., 1997). Therefore, these roots and root holes change water flow behavior (Luo et al., 2019a), further promoting preferential flow in sea buckthorn woodland and natural grassland, consistent with previous findings in the Pisha sandstone area (Liu et al., 2023). In addition, preferential flow in sea buckthorn woodland and natural grassland occurred in shallower soil layer than in bare land. As the forest aged, the soil layer depth at which preferential flow occurred became even shallower (Fig. 4b). This suggested that an increase in roots or root holes may transport water into preferential channels earlier, further enhancing the hydrological connectivity of these channels (Jiang et al., 2018; Yi et al., 2020). Notably, macropores around roots form important water pathways, and root traits play an important role (Luo et al., 2019b). Coarser and larger roots increase infiltration (Vinatier et al., 2024) and preferential flow (Valtera et al., 2023) due to the occurrence of distinct macropores along these roots. Therefore, a stronger well-developed root system can improve and enhance preferential flow more effectively. Indeed, the earlier occurrence of

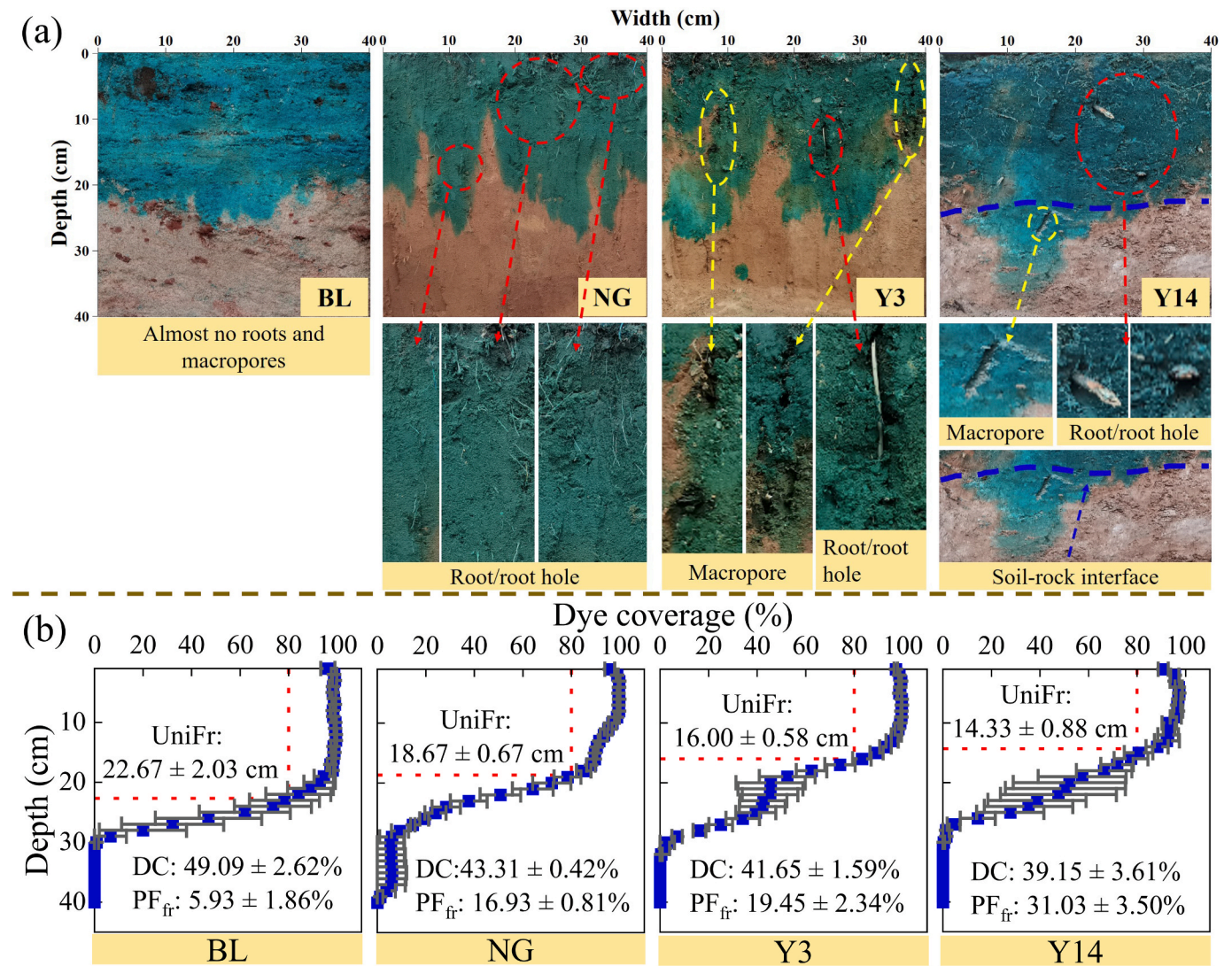


Fig. 4. Dye infiltration patterns (a) and variation of dye coverage with soil depth (b) for bare land (BL), natural grassland (NG) and sea-buckthorn woodland (Y3 and Y14). DC, dye coverage. Unifr, uniform infiltration depth. PF_{fr}, preferential flow fraction. The error bar represents the standard error (n = 3 replications). The red dashed line represents the soil depth corresponding to the dye coverage = 80 % (Unifr). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

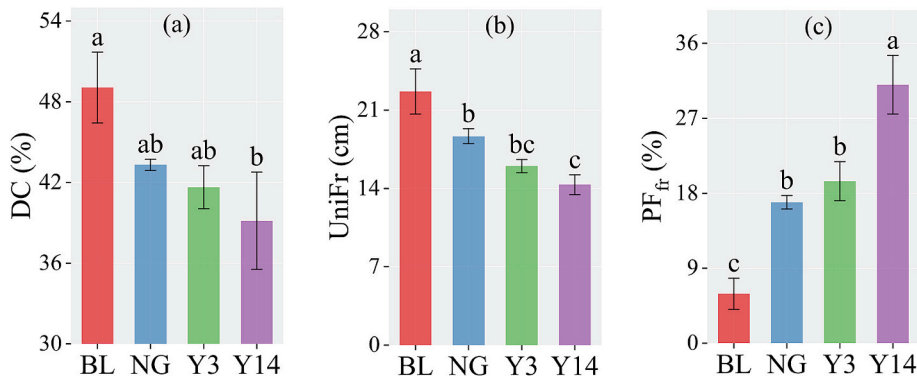


Fig. 5. Characteristic parameters of preferential flow degree for bare land (BL), natural grassland (NG) and sea-buckthorn woodland (Y3 and Y14). DC, dye coverage. Unifr, uniform infiltration depth. PF_{fr}, preferential flow fraction. All data represents the average of three replicates. The error bar represents the standard error. Different lowercase letters represent a significant difference at 0.05 levels (P < 0.05) among different sites.

Table 3
Comprehensive evaluation index (PFI) of soil preferential flow at different plots.

Preferential flow parameters	$E(G_j)$	$\sigma(G_j)$	$W(G_j)$
DC	0.49 ± 0.08	0.37 ± 0.01	0.34 ± 0.00
UniFr	0.57 ± 0.04	0.38 ± 0.01	0.34 ± 0.00
PFFr	0.49 ± 0.01	0.36 ± 0.00	0.32 ± 0.01
Comprehensive evaluation index of preferential flow			
BL	NG	Y3	Y14
$0.03 \pm 0.02c$	$0.43 \pm 0.12b$	$0.70 \pm 0.03a$	$0.90 \pm 0.10a$

Notes: DC, dye coverage. UniFr, uniform infiltration depth. PFFr, preferential flow fraction. BL, bare land. NG, natural grassland. Y3, sea buckthorn woodland aged 3–4 years. Y14, sea buckthorn woodland aged 14 years. Different lowercase letters represent a significant difference at 0.05 levels ($P < 0.05$) among different sites. Values are means $\pm$ standard error ($n = 3$ replications).

preferential flow potentially reduces surface runoff and increases water recharge in arid areas, offering significant ecological benefits (Wang et al., 2020).

Additionally, there were several special preferential flow types in this study (Fig. 4a). “Finger flow,” observed in NG and Y3 plots, and

“funnel flow,” observed in Y14 plots, were largely due to interaction between infiltration water in macropore flow channels and the surrounding soil matrix (Allaire et al., 2009; Weiler and Flühler, 2004). While gravity is crucial in the movement of soil water along root holes or other macropores (Wang et al., 2020), it is not the sole driving force. Matrix suction also contributes by causing water to penetrate into the drier surrounding soil matrix from preferential flow channels (Cey and Rudolph, 2009; Laine-Kaulio et al., 2014). This process, known as “lateral infiltration of macropores flow” (Beven and Clarke, 1986), is common in arid and semi-arid regions with vegetation cover (Wang et al., 2020; Wu et al., 2022). Frequent water exchange between macropores and soil matrix typically occurred near the soil surface (Zhang et al., 2018), where roots and soil pores were more abundant (Fig. 2 and Fig. 3). Therefore, the surface soil (< 20 cm) of natural grassland and sea buckthorn woodland showed more extensive dye coverage compared to the deeper soil (below 20 cm) (Fig. 4). However, dye coverage decreased with soil depth (Fig. 4), indicating reduced water exchange frequency, likely due to decreased root distribution and soil porosity (Wu et al., 2022). In addition, some studies have suggested that “finger flow” is usually observed in sandy soils (Liu et al., 2021; Wang et al., 1998), which aligns with our study area’s high sand content ($> 82\%$) (Table 1).

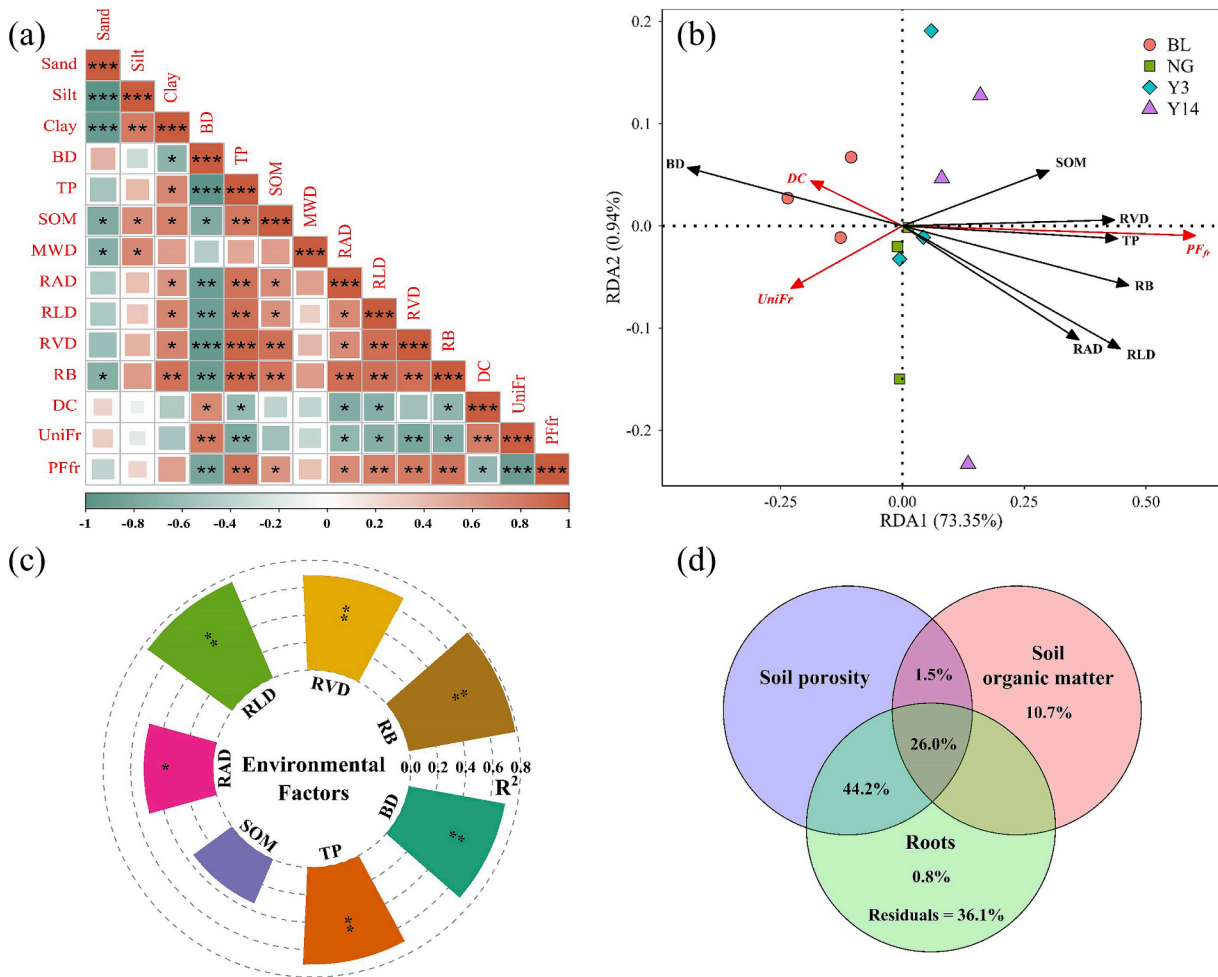


Fig. 6. The correlation between soil properties, root traits, and preferential flow parameters. (a) Pearson correlation matrix analysis between soil properties, root traits, and preferential flow parameters. (b) Redundancy analysis (RDA) used to identify the effects of soil properties and root traits (black arrow) on preferential flow development (red arrow). (c) Environmental importance test analyzed the separate contribution of soil properties and root traits to preferential flow development. (d) Variation-partitioning analysis (VPA) identified the relative contribution of root traits, soil porosity, and soil organic matter to preferential flow development. Root traits include root average diameter (RAD), root length density (RLD), root volume density (RVD), and root biomass (RB). Soil porosity include soil bulk density (BD) and total porosity (TP). Preferential flow parameters include dye coverage (DC), uniform infiltration depth (UniFr), and preferential flow fraction (PFFr). Adjusted R^2 showed the fraction of variation and the shared contribution of each of variables. The intersection area represents the interactive effect between variables. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

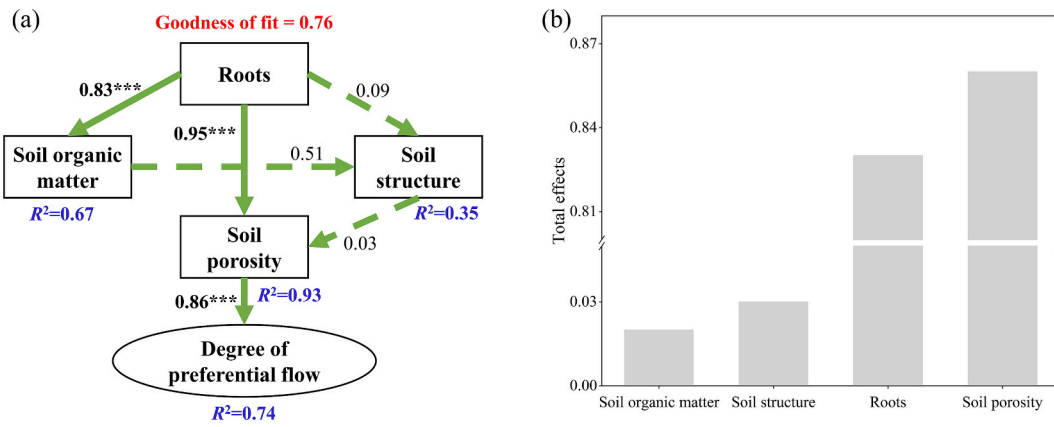


Fig. 7. Partial Least Squares Path Modeling (PLS-PM) showing the effects of root traits, soil organic matter, soil structure, and soil porosity on the degree of preferential flow. (a) and (b) respectively represent the causal paths and total effects of vegetation roots, soil organic matter, soil structure, and soil porosity on the degree of preferential flow. Solid lines represent significant causal paths, while dashed lines represent non-significant causal paths. The number on the arrow indicates the effective normalized path coefficient. R^2 represents the variance of the dependent variable explained by the model. Goodness of Fit statistics are shown above the modeling frames. Root traits include root average diameter (RAD), root length density (RLD), root volume density (RVD) and root biomass (RB). Soil porosity includes total porosity (TP). Soil structure includes soil mean weight diameter (MWD). Degree of preferential flow includes preferential flow fraction (PF_{fr}). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Notably, macropores were observed in shallowly buried soft bedrock in Y14 plots (Fig. 4a), possibly due to vegetation roots embedding into the soft bedrock to access water (Li et al., 2022; McCormick et al., 2021). These root-induced macropores facilitate water diffusion to the bedrock (Luo et al., 2024), potentially increasing water storage in semi-arid areas with shallow soft bedrock. In addition, the “funnel flow” in Y14 may be related to the special soil-rock structure in this area (Fig. 4a), with a thin loess layer overlying the bedrock (Liang et al., 2019). In our study, sedimentary soft rocks were found at a soil depth of 20 cm in Y14 site (Fig. 4a and Fig. S2). There are differences in physical and hydrological properties between bedrock and soil (Yang et al., 2023b), such as variations in saturated hydraulic conductivity ($0.12 \text{ m}\cdot\text{d}^{-1}$, and $0.45 \text{ m}\cdot\text{d}^{-1}$, respectively) and bulk density ($1.54 \text{ mg}\cdot\text{m}^{-3}$, and $1.30 \text{ mg}\cdot\text{m}^{-3}$, respectively), which may lead to different hydrological processes between the two. As upper-layer soil water flowed through sandstone, it moved diagonally down the rock layer into macropores due to the heterogeneous hydraulic characteristics between soil and rock (Nimmo, 2021), such as the lower permeability of sandstone than loess (Barton et al., 2003; Zhang et al., 2020). This formed a funnel-like flow pattern. In fact, in arid and semi-arid regions with shallower bedrock burial, water infiltration patterns may be more complex and diverse. Despite these special flow types identified in the Pisha sandstone area, we emphasized the dominant role of root-hole-induced macropore flow in the infiltration process. This macropore flow could increase the instability of these special flows, enhancing the heterogeneity of infiltration patterns. This effect may grow with the forest age in sea buckthorn woodland, as root traits and total porosity increase over time (Fig. 2 and Fig. 3).

Our results indicated that sea buckthorn woodlands exhibited a higher degree of preferential flow development, followed by natural grassland, and finally bare land (Table 3). This was mainly because sea buckthorn woodlands had a smaller dye coverage, shallower uniform infiltration depth and larger preferential flow fraction and preferential flow index than the other plots (Fig. 5). Previous studies have suggested that vegetation restoration positively impacted soil preferential flow development in loess hilly and gully region, with shrub forests having a greater effect than grassland (Qiu et al., 2023), thereby displaying more preferential flow paths (Bargués Tobella et al., 2014; Kan et al., 2019). Our study further demonstrated that roots and soil pores significantly reduced dye coverage and uniform infiltration depth while increasing preferential flow fraction and index (Fig. 6a). This suggested that roots and soil pores positively influenced preferential flow development,

explained for 71.0 % and 71.7 % of the variation in preferential flow, respectively (Fig. 6c). Good root conditions promote stable macropore formation under the forest, enhancing water preferential flow (Tang et al., 2019). In addition, root decomposition or root secretion could increase soil organic matter content and improve soil structure, further supporting preferential flow development (Zhang et al., 2016). Therefore, vegetation restoration can improve soil porosity by promoting root growth, which in turn enhances soil water penetration and preferential flow development (Bargués Tobella et al., 2014; Huang et al., 2017; Jiang et al., 2019). This was well verified in our study (Fig. 7). In conclusion, vegetation restoration induced soil preferential flow, changed soil water infiltration pattern, and the roots and root holes significantly improved preferential flow development in the Pisha sandstone area.

4.3. Hydrological effects of vegetation restoration on areas with shallow bedrock burial

The Pisha sandstone area is influenced by regional tectonic movements, resulting in a thin layer of overlying soil and shallow bedrock that is even exposed at the surface (Liang et al., 2019). Once this area is affected by drought, the water in the shallow soil may evaporate quickly. The underlying low-permeability sedimentary rock further hinders rainfall infiltration, making it difficult to collect and store rainwater in arid areas (Qin et al., 2015; Rodríguez-Robles et al., 2015; Yang et al., 2023b). Therefore, regions with shallow bedrock may rely entirely on rainfall to provide moisture for vegetation, posing challenges for the long-term survival of plant life. However, existing research indicates that once the shallow soil becomes dry, vegetation may alter its water uptake strategy, subsequently sourcing moisture from weathered bedrock and deep water associated with rock fractures (Querejeta et al., 2007; Schwinning, 2008; Poot et al., 2012). In this study, the macropores induced by roots observed in the stained profiles also indicated the historical penetration of plant roots into the sandstone to access moisture (Fig. 4a). Additionally, the macropore created by root growth and decay facilitate the infiltration of water into the underlying sedimentary rock, partially overcoming the challenges posed by low-permeability sedimentary layers. Thus, vegetation restoration can enhance soil permeability by improving soil structure and increasing soil porosity, thereby promoting the infiltration of moisture into the deeper sedimentary rock. However, these challenges are not unique to our study area; similar geological conditions can be found worldwide, such as in

the northern Caspian Plain (Badyukova, 2018), the Arkhangelsk area in northwest Russia (Lyså et al., 2001), karst regions (Qin et al., 2015), the Tibetan Plateau in China (Wang et al., 2023), and the Barua watershed in the northern foreland of the Peninsula Indian (Yadav et al., 2020). Understanding how these conditions affect water infiltration and the impact of different vegetation types on the hydrological processes in areas with shallow bedrock can provide valuable insights for vegetation restoration and water resource management in similar environments.

4.4. Ecological implications and limitations of the present study

Enhancing soil moisture is crucial for vegetation growth in arid and semi-arid regions, making it an essential component of vegetation restoration and ecological revitalization (Liu et al., 2020). Understanding the effects of different vegetation restoration types on soil properties, water infiltration, and preferential flow development is essential for formulating effective regional vegetation restoration strategies. Our study confirmed that the roots and soil pores in sea buckthorn woodland and natural grassland altered infiltration patterns and promoted preferential flow development in the Pisha sandstone area with shallow bedrock burial. Root-facilitated macropores can enhance soil aeration, thereby creating favorable conditions for vegetation growth (Luo et al., 2010). We observed a positive correlation between the preferential flow index and aboveground herbaceous biomass (or litter accumulation) (Fig. 8), highlighting the importance of preferential flow development in fostering aboveground vegetation growth (nutrient cycling). Therefore, the increasing the degree of soil preferential flow can significantly enhance water infiltration, enabling the rapid transport of water and nutrients essential for sustaining vegetation restoration in arid and semi-arid areas (Wu et al., 2022). In addition, the study revealed that sea buckthorn woodland, especially the Y14, exhibited more developed preferential flow compared to natural grassland. Consequently, sea buckthorn woodland may have greater potential to alleviate water restriction in shallow bedrock areas with challenging water infiltration. Furthermore, it also could serve as a more efficient short-term strategy for vegetation restoration to mitigate water stress in drylands. Importantly, the results of this study are not limited to the arid and semi-arid areas in northern China and could serve as a reference for addressing similar challenges in other arid and semi-arid regions worldwide.

This study confirmed the impact of different vegetation types on water infiltration patterns and preferential flow, but it did not address their long-term effects. Further exploration is necessary to fully understand the hydrological regulation mechanisms of vegetation restoration in arid and semi-arid areas with shallow bedrock. While Brilliant Blue FCF dye tracer is a widely used tracer, it also exhibits some deficiencies, like problematic sorption and lower mobility than pure water in small pores (Nobles et al., 2010; Wang and Zhang, 2011). To more accurately simulate water transport in soil, using a combination of tracers, such as Brilliant Blue and Br^- , is recommended. In addition, although the dye tracer experiment is a convenient and intuitive method for identifying infiltration patterns and preferential flow, its results are based on two-dimensional static characterization. Future studies should incorporate X-ray computed tomography (CT) to examine the three-dimensional characteristics of soil structure, providing a more detailed description of water movement in soil.

5. Conclusions

Vegetation restoration could improve soil nutrient level and structure by promoting root development, which in turn facilitated the creation and progression of soil preferential flow in Pisha sandstone area. Vegetation roots and soil pores were the principal factors affecting preferential flow development, accounting for 71.0 % and 71.7 % of the changes of preferential flow, respectively. However, the effects of different vegetation types on soil properties and preferential flow varied. Sea buckthorn woodland, compared to natural grassland, exhibited a

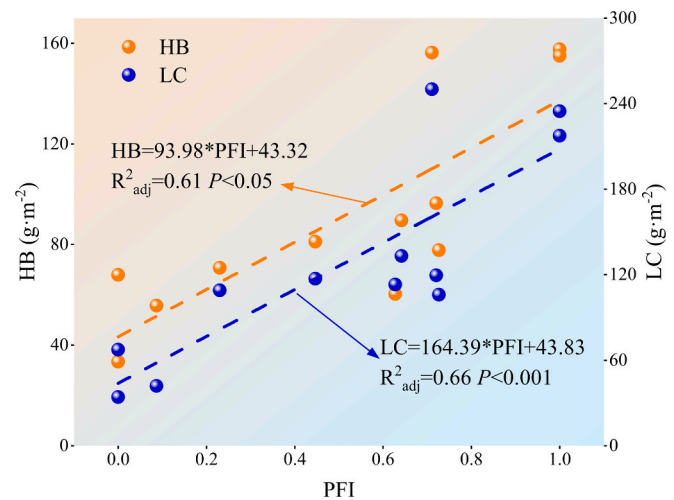


Fig. 8. Linear regression of aboveground herbaceous biomass and litter accumulation to preferential flow index. HB, aboveground herbaceous biomass; LC, litter accumulation; PFI, preferential flow index.

superior influence, with the degree of improvement increasing as the sea buckthorn forest aged. Moreover, the development of preferential flow further fostered herbaceous growth and litter accumulation. Given the beneficial role of preferential flow in enhancing water infiltration, vegetation growth, and nutrient cycling in arid regions, it is recommended that local authorities and farmers prioritize planting sea buckthorn to alleviate water resource constraints and vegetation scarcity in the Pisha sandstone area characterized by shallow bedrock burial. The results provide new insights into the effects of vegetation restoration on hydrological processes in arid and semi-arid regions with shallow bedrock. Additionally, these results offer guidance for water resources management and ecological restoration in drylands with challenging infiltration conditions.

CRediT authorship contribution statement

Jian Hu: Writing – review & editing, Writing – original draft, Validation, Investigation, Conceptualization. **Yunzhuo Ren:** Resources, Investigation. **Mei Tang:** Resources, Investigation. **Zhiao Zhang:** Resources, Investigation. **Kaiqi Yang:** Resources, Investigation. **Qing Zhen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Fengpeng Han:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhyd.2025.133546>.

org/10.1016/j.jhydrol.2025.133546.

Data availability

Data will be made available on request.

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